



The limits of the inertia bound for graph powers

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Inertia and graphs

Definition

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An **independent set** in a simple graph G is a set of pairwise non-adjacent vertices.

$$\alpha(G) := \max\{|S| \mid S \text{ is an independent set in } G\}.$$

Inertia and graphs

Inertia bound (Cvetković '71, Calderbank-Frankl '92)

Let A be a weighted adjacency matrix of G , then

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We write

$$n_{\geq 0}(G) := \min_A n_{\geq 0}(A),$$

where the minimum runs over all weighted adjacency matrices of G .

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No! (Sinkovic 2018)

Sometimes a weak bound

Theorem (Kwan-Wigderson 2024)

There is an infinite family of C_4 -free graphs $(G_n)_n$ with

$$\alpha(G_n) = O(n^{3/4}) \quad \text{and} \quad n_{\geq 0}(G_n) \geq .232n.$$

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$$\alpha(H_n) \leq 2 \quad \text{and} \quad n_{\geq 0}(G_n) = \Omega(n^{1/9}).$$

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If the complement of G contains a spanning (n, d, λ) graph with $d < n/2 - 1$, then

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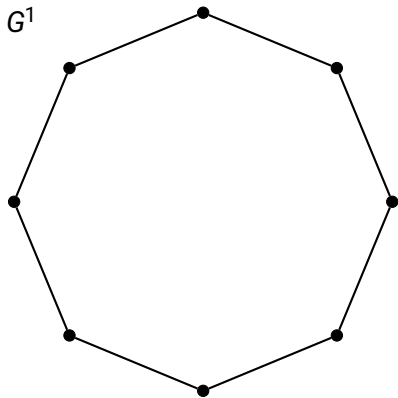
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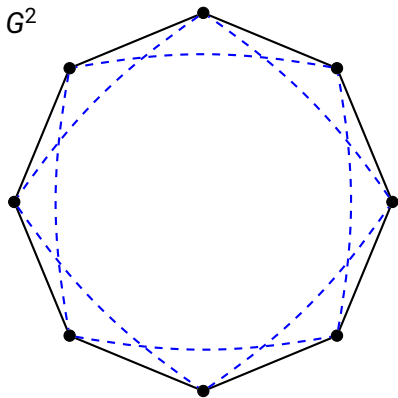
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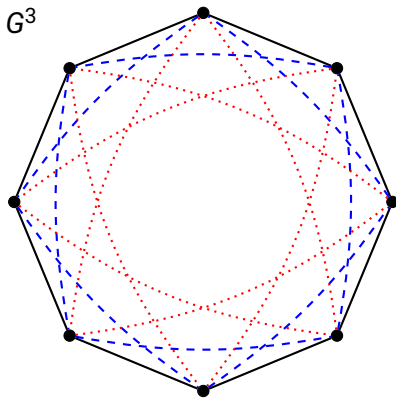
Graph powers



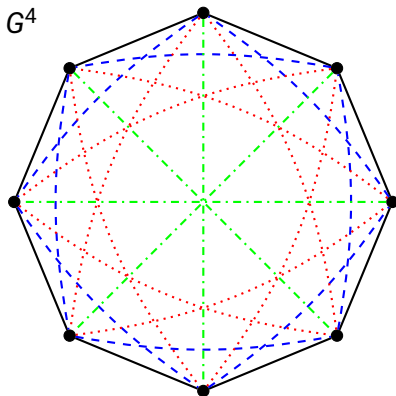
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Graph powers

$$\alpha(G)$$

size of the largest set of vertices at pairwise distance at least 2.

$$\alpha(G^k)$$

size of the largest set of vertices at pairwise distance at least $k + 1$.

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Cayley graphs $\text{Cay}(\Gamma, A)$

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If $|S| = O(\sqrt{N/\log N})$, then there exists $T \subset \mathbb{Z}/N\mathbb{Z}$ such that $T + T = \mathbb{Z}/N\mathbb{Z} \setminus S$.

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If S is a random subset of $\mathbb{Z}/N\mathbb{Z}$ then $\bar{G}$ is with high probability an $(N, |S|, \sqrt{|S| \log n})$ -graph.

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With high probability $x + y + z = 0$ has no solutions in S .
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Theorem (Abiad-M.-van de Berg 202?)

There is family of power graphs G_n and $\delta > 0$ such that

$$\alpha(G_n^2) \leq 2 \quad \text{and} \quad n_{\geq 0}(G_n) = \Omega(n^\delta).$$



Thank you for your attention!

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